



About the workshop

AI Technologies are going to dominate this planet's future. Optimization methods play a critical role in developing and training AI models. The objective of an AI model needs to be cast as an optimization problem that requires efficient solutions for effective and accurate decision-making.

This workshop is designed to provide an overview of various optimization techniques commonly used in developing AI models. Participants will be able to work through hands-on exercises (Python/Matlab) and capstone projects to understand the development of AI/ML/DS models through optimization glass. By the end of the workshop, participants will have a detailed understanding of AI/ML/DS models and different optimization algorithms used for their training and development.

Objectives of the workshop

The five-day workshop and FDP on “Optimization Techniques for AI/ML/DS Models” focus on an overview of optimization techniques and their various applications in AI domains. The main objectives of this workshop are:

- To understand the significance of optimization in AI.
- To understand various optimization techniques and their applications.
- To provide hands-on sessions for a better understanding of these concepts.
- To explore various research opportunities and challenges in the fields of AI and optimization.



Topics covered

- Introduction of optimization and its significance in AI
- Mathematical foundation for optimization
- Gradient descent algorithms: Strength, weakness and applications
- Regression and classification models: their optimization frameworks and solutions
- Optimization algorithms for Deep learning models
- Meta-heuristic algorithms: functionality and applications



Five-day workshop and FDP on Optimization Techniques for AI/ML/DS Models

MAY 22-26, 2023

**Department of Computational Science
and Humanities**

www.iiitkottayam.ac.in

ABOUT THE INSTITUTE

The Indian Institute of Information Technology, Kottayam is one among the IIITs that has been established as "Institutions of National Importance" by the Ministry of Education, Govt. of India under the ambit of IIIT (PPP) Act 2017. The institute offers 4 year B.Tech and B.Tech(Hons) program with specialized courses covering all major research domains of Computer Science and Electronics and Communication Engineering. The students are admitted against the JoSAA/CSAB counselling against the JEE main rank list. The institute also offers MTech in CSE and Ph.D. programs in CSE, ECE and Mathematics.



CONTACT

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ABOUT THE DEPARTMENT

The Department of Computational Science and Humanities was formally established in 2019 with an aim to develop mathematically sound young minds for solving industry-related problems arising from different industries and related areas. The department consists of seven full-time faculty members and 3 joint faculty members. It offers core courses in Mathematics/Statistics/Humanities to all B.Tech., M.Tech, and Ph.D. programs. This department takes pride in offering an exclusive doctoral program in Mathematics, meticulously crafted to meet the requirements of highly motivated and aspiring graduates holding a degree in this field.

RESOURCE PERSONS

- Dr. Jayakrushna Sahoo, IIITK
- Dr. Dhanyamol M V, IIITK
- Dr. Susheel Kumar Joshi, IIITK
- Dr. Pritam Anand, Dhirubhai Ambani Institute of Information and Communication Technology, Ahmedabad, Gujarat
- Dr. P.Victor Paul, IIITK
- Dr. J.V. Bibal Benifa, IIITK
- Dr. John Paul Martin, IIITK
- Dr. Christina Terese Joseph, IIITK
- Dr. Keshab Nath, IIITK
- Dr. Nilotpal Chakraborty, IIITK
- Dr. Rubell Marion Lincy G, IIITK
- Dr. Manu Madhavan, IIITK

PATRONS

- Prof. (Dr.) Rajiv Dharaskar, Director, IIIT Kottayam.
- Dr. M.Radhakrishnan, Registrar, IIIT Kottayam.
- Prof. (Dr.) P. Mohanan, Prof. In-Charge (Academics), IIIT Kottayam.

PROGRAM COORDINATORS

- Dr. T. Bakkyaraj, Assistant Professor and Head.
- Dr. Susheel Kumar Joshi, Assistant Professor.
- Dr. Dhanyamol M V, Assistant Professor.
Department of Computational Science and Humanities, IIIT Kottayam

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1. Choose the category as Educational Institutions
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REGISTRATION

Target Audience: Faculty, PhD scholars, UG/PG students and Industry Personnel.

Registration fee: INR 1500/- (including GST)

Accommodation (on a shared basis): Provided upon request via email.

Registration deadline: 19/05/2023

(Seats will be assigned on a first-come, first-served basis due to limited availability.)

Registration link: <https://tinyurl.com/bp725796>

ORGANISING TEAM

- Dr. Rubell Marion Lincy G
- Dr. Balu A
- Dr. Asha Sebastian
- Dr. Murugan D
- Dr. Gayathri G R
- Dr. Josit Mariya
- Dr. Mathew C D
- Mr Anandhu Mohan
- Mrs. Reetha Thomas
- Ms. Nicky Caroline S
- Ms. Julie Sara Benny
- Mrs. Ansmol George
- Mrs. Sheba Elizabeth Thomas
- Ms. Afla A S
- Ms. Fathima Nihla Latheef